「Preliminary.

「Lemma. Let $T: V \to V$ be a linear map on finite-dimensional space V, and let $W \subseteq V$ be a T-invariant subspace.
Suppose $v_1, \cdots, v_k$ are eigenvectors of T corresponding to distinct eigenvalues.
If $v_1 + v_2 + \cdots + v_k \in W$, then $v_i \in W$ for all $1 \le i \le k$

proof. If $k=1$, trivial. Suppose that for any integer smaller than k, the statement holds.
Assume $v_1, \cdots, v_k$ are eigenvectors of T corresponding to distinct eigenvalues λ_i.
If $v_1 + \cdots + v_k \in W$, then

$$\underbrace{T(v_1 + \cdots + v_k) = \lambda_1 v_1 + \cdots + \lambda_k v_k \in W}_{W \text{ is } T\text{-invariant}} \quad \text{and} \quad \underbrace{\lambda_k (v_1 + \cdots + v_k) \in W}_{\text{Closed under scalar multiplication}}$$

Subtracting two vectors:

$$\sum_{i=1}^{k} (\lambda_i - \lambda_k) \cdot v_i = \sum_{i=1}^{k-1} \underbrace{(\lambda_i - \lambda_k)}_{i \neq k \Rightarrow \text{not zero}} \cdot v_i \in W$$

Therefore, $v_i \in W$ for all $1 \le i \le k-1$, by the hypothesis. Moreover,

$$v_k = v_1 + \cdots + v_k - (v_1 + \cdots + v_{k-1}) \in W.$$

「Thm. If $T: V \to V$ is diagonalizable, then for any T-invariant $W \le V$,

$$T|_W : W \to W : x \mapsto T(x)$$

is also diagonalizable.

proof. Let β be a basis for V s.t. $[T]_\beta$ is diagonal.
Let β' be a basis for W. Then, every vector of β' is represented by linear combination of β.
That is, if $\beta = \{v_1, \cdots, v_n\}$ and $\beta' = \{w_1, \cdots, w_k\}$, then for all $1 \le j \le k$,

$$w_j = a_{1,j} v_1 + \cdots + a_{n,j} v_n \quad (a_{ij} \in F).$$

Consider a n by k matrix $F^k \to F^n$

$$\begin{pmatrix} [w_1]_\beta & \cdots & [w_k]_\beta \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ a_{n1} & - & - & a_{nk} \end{pmatrix}$$

has rank as k.

「Simultaneously Diagonalizable.

In this section, V is finite-dimensional vector space over F, and $T_1, T_2, \cdots : V \to V$ are linear operator.

「Def. Linear operators T_1 and T_2 are called simultaneously diagonalizable if:

there exists a basis β for V s.t both $[T_1]_\beta$ and $[T_2]_\beta$ are diagonal.

「Lemma. If T_1, T_2 are simultaneously diagonalizable by a basis β, then for any basis α, $[T_1]_\alpha$ and $[T_2]_\alpha$ are simul. diagonalizable.

Proof. Clearly, $[T_i]_\beta = [T_i]_\alpha^\beta [T_i]_\alpha [T_i]_\beta^\alpha$ thus obtain the result.

「Lemma. If T is diagonalizable, then T^m and T are simultaneously diagonalizable, for any $m \geq 1$

Proof. Let β be a basis for V satisfying $[T]_\beta$ is diagonal. Then

$$[T^m]_\beta = [T]_\beta^m \quad \text{diagonal.}$$

「Thm. If T_1, T_2 are simultaneously diagonalizable by a basis β, then T_1 and T_2 commute. That is, $T_1 \circ T_2 = T_2 \circ T_1$

Proof. $[T_1 \circ T_2]_\beta = [T_1]_\beta \cdot [T_2]_\beta = [T_2]_\beta \cdot [T_1]_\beta = [T_2 \circ T_1]_\beta.$

$\Updownarrow$ Converse

$$\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \cdot \begin{pmatrix} \partial_1 & & 0 \\ & \ddots & \\ 0 & & \partial_n \end{pmatrix} = \begin{pmatrix} \lambda_1\partial_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n\partial_n \end{pmatrix}$$

「Thm. If T_1 and T_2 are diagonalizable and commute, then T_1 and T_2 are simultaneously diagonalizable.

Thm. Let C be a collection of diagonalizable linear operator on V.

there exists a basis β for V s.t $[T]_\beta$ is diagonal for all $T \in C$

if and only if

C commute under composition.